



Chemical and electrochemical intercalation of lithium into boronated carbons

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Abstract

Boron-substituted carbons have been produced by chemical vapour deposition from acetylene and boron trichloride precursors at 1140°C. Li intercalation was investigated, chemically from the Li vapour and electrochemically in Li–carbon cells. In both cases, the amount of intercalated Li increases with the boron content of the carbon, up to a value of 13 at.% boron. Above this value, the intercalation of lithium is not so efficient. This behaviour is understood by assuming that boron, which acts as an electron acceptor and hence favours the intercalation of electron donors like lithium, can enter substitutionally into the carbon lattice up to a certain limit, which is close to 13 at.% for the materials deposited at 1140°C. © 1999 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Boron is one of the few elements which are known with some certainty to enter substitutionally into the carbon lattice. Chemical reactions at, or near, thermodynamic equilibrium lead to low boron content materials. Lowell [1] has shown that the thermodynamic solubility of boron in the carbon lattice cannot exceed 2.35 at.% and Marchand [2] summarized various techniques for doping graphitic carbons and graphites with boron (0.1–1 at.%). More recently, several groups [3,4] have shown that substantial increases in substitutional boron, up to a composition close to BC_3 , can be obtained by chemical vapour deposition (CVD) from BCl_3 and a hydrocarbon at temperatures between 800 and 1100°C, although it is not sure that all boron atoms are in substitution for the highest concentrations [5].

As boron has one less electron than carbon, it acts as electron acceptor and therefore modifies strongly the electronic properties without any large distortion of the crystal lattice. Thus, a change in intercalation behaviour is expected compared to carbons. A few studies have been devoted to the intercalation chemistry of these materials [6–8]. It was shown that electron acceptors, such as

halogens, do not react with materials of high boron content, whereas intercalation of electron donors (alkali metals) is easier than in graphite. Of particular interest is the intercalation of lithium from the vapour phase: whereas lithium carbide is formed with pristine graphite at temperatures higher than 450°C, such a reaction did not seem to occur with BC_3 , even at 600°C, where high intercalated Li contents were obtained [7], twice more than in graphite. In this paper, we examine the chemical and electrochemical intercalation of lithium into boronated carbons prepared by CVD.

2. Experimental

Boron substituted carbons were synthesized by CVD from acetylene and boron trichloride as precursors and hydrogen as dilution and carrier gas. Boronated films were deposited on quartz substrates placed in an alumina hot-wall reactor maintained at 1140°C. The compositions of the deposits were determined by electron probe micro-analysis. Further detail on preparation and characterization of the samples can be found elsewhere [5,9]. One of the main results is the increase in crystallinity with the boron content up to a limit which is of the order of 13 at.% for the deposition temperature of 1140°C. Then for higher boron concentrations, the crystallinity decreases, as shown

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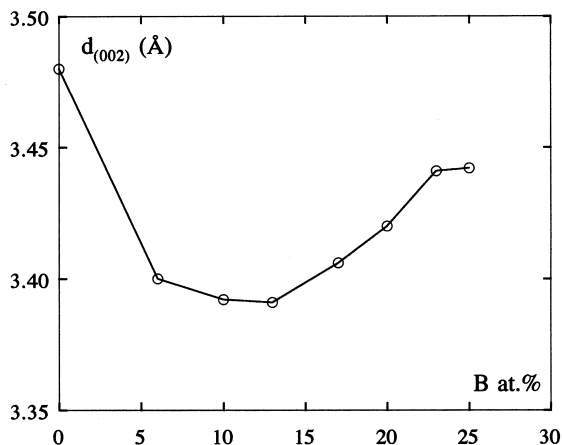


Fig. 1. The dependence of the interplane distance d_{002} on the boron content of films deposited at 1140°C.

in Fig. 1 for the dependence of the interplane distance d_{002} on the boron content.

Lithium intercalation into pieces of the films removed from the substrate was carried out from lithium vapour using stainless-steel reactors. The composition of the intercalated samples was estimated from weight uptake measurements and the repeat distance along the c -axis was deduced from X-ray diffraction patterns.

Electrochemical intercalation was performed in Li–C cells with metallic lithium as counter electrode and microporous propylene sheet (Celgard) as separator. The carbon electrodes were prepared by mixing carbon powders (90%) obtained from the films ground in a mortar, teflon (5%) and acetylene black (5%). The electrolyte was LiPF_6 (1 M) dissolved in EC/PC/3DMC. The cells were cycled using a Mac Pile galvanostat at constant current intensity (5 mA g^{-1} of boronated carbon).

3. Results and discussion

3.1. Li intercalation from the vapour phase

Fig. 2 shows the mass uptake as a function of the reaction duration at two reaction temperatures (400 and 500°C) for a boronated sample of composition $\text{B}_{0.13}\text{C}_{0.87}$. This composition corresponds to the minimum of d_{002} value (3.39 Å) at the deposition temperature of 1140°C (Fig. 1). For both intercalation temperatures, a Li uptake limit is reached after 24 h reaction. X-ray diffraction spectra (Fig. 3) show that a stage-1 compound is obtained, with a mean repeat distance I_c of about 3.75 Å, as expected from previous work on Li intercalation into graphite [10]. For shorter reaction duration, peaks characteristic of stage-2 and stage-3 are clearly observed. In the case of the reaction at 500°C, the limit of Li uptake corresponds to the composition $\text{Li}(\text{B}_{0.13}\text{C}_{0.87})_{3.8}$, higher

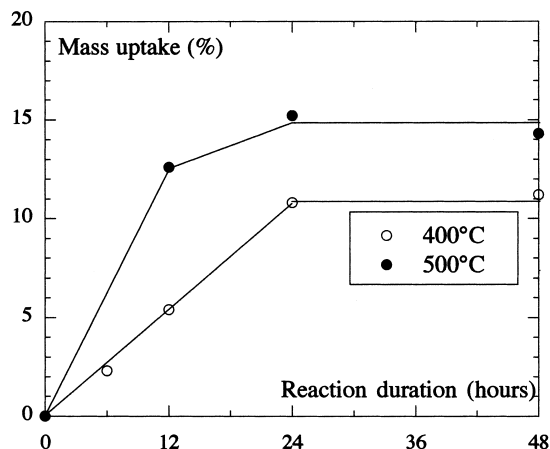


Fig. 2. Li intercalation from the vapour phase into $\text{B}_{0.13}\text{C}_{0.87}$ samples as a function of the reaction duration at 400 and 500°C.

than the limit in graphite (LiC_6). At higher temperatures of reaction, some lithium carbide begins to appear, as was seen by X-ray diffraction.

Fig. 4 shows the composition limit as a function of the boron content for the two reaction temperatures of 400°C and 500°C. Both curves exhibit a maximum for 13 at.%

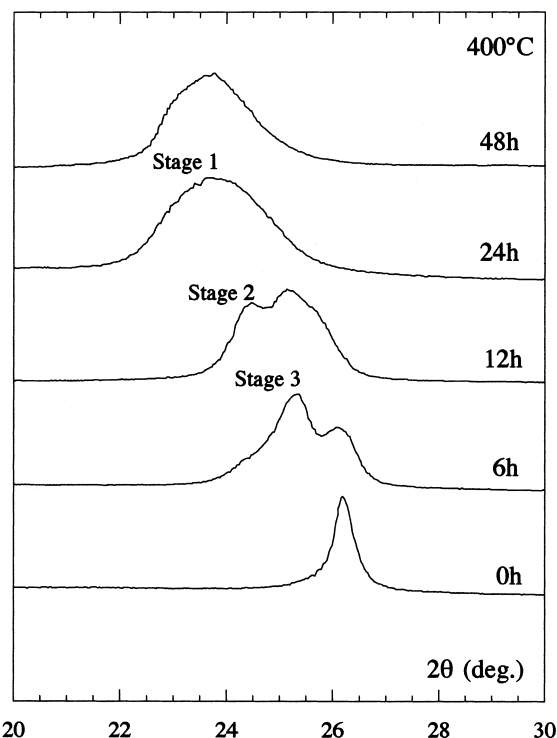


Fig. 3. X-ray diffraction patterns ($\text{CuK}\alpha$) of Li intercalation compounds prepared from $\text{B}_{0.13}\text{C}_{0.87}$ after exposure to Li vapour for different lengths of time.

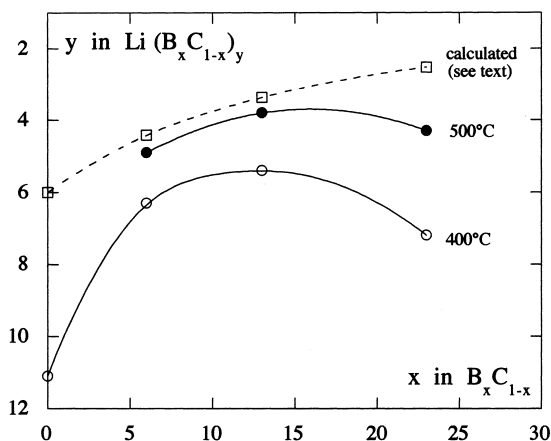


Fig. 4. The maximum amount of lithium intercalated from the vapour phase as a function of the boron content.

boron, the boron content which corresponds to the highest crystallinity of the host material. In this figure, the experimental results are compared to the expected maximum of intercalated lithium (dotted curve), which was calculated from the excess Li with respect to LiC_6 , needed to compensate for the loss of electron due to the boron content. Assuming a quasi-rigid band system and a loss of one electron per boron atom, the y value in a compound of composition $\text{Li}(\text{B}_x\text{C}_{1-x})_y$ is then $y=6/(6x+1)$. Clearly, the experimental values obtained at 500°C are close to the calculated ones, except for boron contents higher than 13 at.%.

The decrease in intercalated Li amount for boron contents higher than 13 at.% leads to the conclusion that boron atoms incorporated in the carbon lattice above this limit do not act as electron acceptors. The most simple explanation is that they are located in other sites than in substitution. One can imagine interstitial sites between the carbon planes, where isolated boron atoms or small boron clusters could be located, or, eventually, formation of boron carbide, although X-ray diffraction did not show evidence for the latter compound. This would produce an increase in d_{002} as shown in Fig. 1. Now, if we assume that a composition $\text{B}_{0.23}\text{C}_{0.77}$ can be written as $0.115\text{B} + 0.885(\text{B}_{0.13}\text{C}_{0.87})$, the y value expected for $\text{Li}(\text{B}_{0.23}\text{C}_{0.77})_y$ would be $y=3.8/0.885=4.29$ at 500°C and $y=5.4/0.885=6.1$ at 400°C, close to the experimental values (4.3 and 7.2, respectively; see Fig. 4). This result seems to show that the substitution of carbon atoms by boron is limited to a value of 13 at.% for a deposition temperature of 1140°C.

3.2. Electrochemical intercalation

Fig. 5 shows the first discharge and charge of Li cells with electrodes made up of carbons deposited at 1140°C containing 0 and 13 at.% boron, respectively. During the

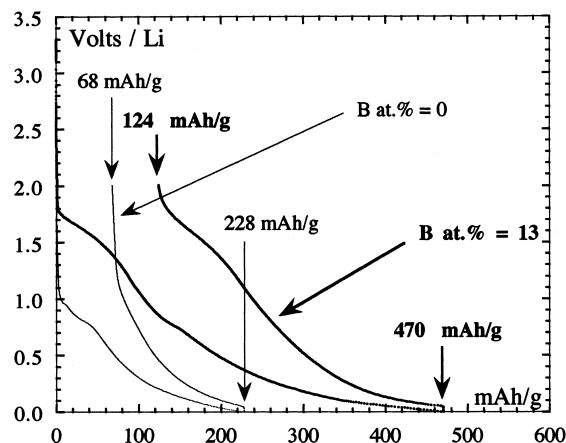


Fig. 5. First discharge and charge of Li-carbon cells with and without substitutional boron.

first discharge, electrolyte decomposition occurs on the surface of the carbon, leading to the formation of a passivating layer [11]. This decomposition consumes Li, leading to irreversible losses during the first cycle, the so-called irreversible capacity. It occurs generally at a potential close to 0.8 V vs. Li^+/Li , producing a voltage plateau clearly visible on the curves of Fig. 5. Another voltage plateau, at higher potential (1.5–1.6 V), appears on the discharge curve corresponding to the boron-containing electrode. This plateau is reversible, at least partly, since it also appears on the charging curve. Its origin is not well understood. It was also observed in a previous study [8] of the electrochemical intercalation of lithium into boronated carbons produced by CVD from benzene and boron trichloride at 900°C.

The voltage profiles did not exhibit any other plateau at low potential. This was expected for the pure carbon deposited at 1140°C, whose crystallinity is too poor for the staging phenomena to occur [12], but not for the $\text{B}_{0.13}\text{C}_{0.87}$ sample, which was shown to give successively stage-3, stage-2 and stage-1 by chemical intercalation (see Section 3.1). Voltage plateaus corresponding to transitions between these stages could be expected.

The amount of lithium which can be recovered on charge corresponds to the reversible capacity of the cells. The voltage curves of Fig. 5 show that the cell with the boron-containing electrode has a reversible capacity (346 mA h g^{-1}) more than twice larger than the cell without boron (160 mA h g^{-1}). However, part of the capacity occurs above 1 V vs. Li^+/Li . This behaviour can be understood qualitatively by taking into account the acceptor character of boron. The presence of boron strengthens the chemical bond between the intercalated Li and the boron-carbon host compared to the pure carbon host [8]. As a result, the potential is increased relative to unsubstituted carbon.

From the Li uptake obtained by chemical intercalation

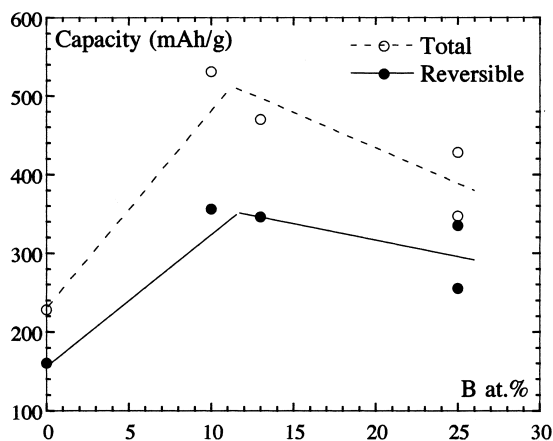


Fig. 6. The capacities of Li-carbon cells as a function of the boron content.

one would expect reversible capacity values of about 200 mA h g^{-1} for the unsubstituted carbon and above 500 mA h g^{-1} for the 13 at.% boronated sample. The lower values given by electrochemical intercalation, especially for the boron containing sample, are not well understood. Nevertheless, the same trend is observed (Fig. 6) for the evolution of the capacity as a function of the boron content.

4. Conclusion

Due to its acceptor character, substitutional boron causes an increase in the amount of lithium intercalated chemical-

ly or electrochemically, compared to unsubstituted carbon. However, a maximum of Li uptake is observed as a function of the boron content. It occurs for about 13 at.% boron in the materials deposited at 1140°C . This seems to indicate that there exists a limit to boron substitution in carbon, which depends on the CVD conditions. This result is in agreement with other characterization studies [9] performed on boronated carbons produced by CVD.

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